01 / THE DEVELOPMENT

Clock-Free Scale Compass

Three normalized readings, one held-out prediction,
and an exact comparison circle

The previous physics-cubit construction retained unknown physical calibrations as native comparison records. This development adds an inverse observation protocol: recover both positive transport parameters from measured invariant ratios, locate their relative balance point, and predict an unused setting.

For the inherited two-generator cell, write

$$\chi(q,p) = \frac{16q^2p^2}{(1+q^2)^2(1+p^2)^2}, \qquad q,p>0.$$

The declared comparison operation changes a length ratio s and a source-energy ratio t independently:

$$q(s,t) = stq, \qquad p(s,t) = tp/s.$$

Both ratios must be known independently of the invariant being fitted. During the comparisons, the transport calibration is held fixed and departures from exact cut compatibility are retained.

Main result. Three normalized readings at $(s,t) = (1,1), (2,1), (1,2)$ uniquely recover positive q,p whenever they satisfy this model. They predict the fourth reading at $(2,2)$ with no additional fit. After centering the two relative scales, the same model obeys

$$\boxed{\frac{1}{\sqrt{\chi}} - 1 = X^2 + Z^2.}$$

Here X, Z are comparison coordinates, defined on page 6. The circle is an exact geometry of these coordinates; its interpretation as physical spacetime has not been established.

What is new, and what is inherited

Inherited core	Native cut arithmetic, Cayley transports, the loop defect, the nontrivial compatible cell $q = p = 1$, and stationary series response.
New normalization	Eliminate an unknown common response gain while retaining the relative link-weight profile.
New obstruction	The entire length-only scalar response, spectrum and rooted resolvent cannot distinguish qp from $(qp)^{-1}$.
New reconstruction	Add an independent source-energy comparison; recover both ratios and test a fourth setting.
New geometry	Derive the unique relative balance point and its exact circular level sets.
Verification	10 native symbolic certificates, 10 serialized replays, 610 exact checks and 10 negative controls.

These are results inside a declared finite native model. The examples are exact synthetic computations, not experimental observations of gravity. Physical reference readings remain necessary to assign the recovered parameters to dimensional constants.

02 / SCALE RECORDS

Remove the common gain

C1. Weighted-cycle normalization. Let a simple cycle with $m \geq 3$ links carry unitary native transports. Its loop H satisfies

$$(H - \mathbf{1})^\dagger (H - \mathbf{1}) = 4\chi \mathbf{1}_2.$$

Assign positive link weights $w_e = g\bar{w}_e$, with unknown common gain $g > 0$ and known relative weights $\bar{w}_e$. At a chosen root, let K_{bare} be its diagonal stiffness block with the other coordinates held fixed. Let K_{eff} be the root Schur operator after stationary elimination of the interior coordinates. Put

$$\bar{R} = \sum_{e=1}^{m} \frac{1}{\bar{w}_e}, \qquad \bar{b} = \bar{w}_1 + \bar{w}_m.$$

Then

$$K_{\mathrm{bare}} = g\bar{b}\mathbf{1}_2, \qquad K_{\mathrm{eff}} = \frac{4g\chi}{\bar{R}}\mathbf{1}_2,$$

and the gain cancels:

$$\boxed{\chi = \mathcal{N}\frac{\operatorname{tr} K_{\mathrm{eff}}}{\operatorname{tr} K_{\mathrm{bare}}}, \qquad \mathcal{N} = \frac{\bar{R}\bar{b}}{4}.}$$

Proof. Write the quadratic cost as $\frac{1}{2}\sum_e w_e \|x_{e+1} - T_e x_e\|^2$. Gauge away the transports on an opened path. For a fixed root vector x, the transported increments d_e have total mismatch Δ, with $\|\Delta\|^2 = 4\chi\|x\|^2$. Minimizing $\frac{1}{2}\sum_e w_e \|d_e\|^2$ subject to $\sum_e d_e = \Delta$ gives

$$d_e = \frac{w_e^{-1}}{\sum_f w_f^{-1}}\Delta, \qquad \min Q = \frac{\|\Delta\|^2}{2\sum_e w_e^{-1}}.$$

This is precisely the Schur quadratic form, yielding K_{eff}. The bare root receives the two incident weights. Taking traces proves the ratio. □

This uses algebraic stationary elimination. A claim that a physical system relaxes to that stationary state would require its own dynamics and detector adapter.

Refinement changes the normalizer

Relative link weights	$\bar{R}$	$\bar{b}$	$\mathcal{N}$	Result
Four links: $(1,1,1,1)$	4	2	2	Same χ
Four links: $(1,2,3,4)$	25/12	5	125/48	Same χ
Eight links: all 2	4	4	4	Same χ

In the last row each original transport T is split into $(\mathbf{1}, T)$, with two relative weights 2. The series response and loop are preserved. Reusing the four-link normalizer 2 would give a wrong invariant.

Observation contract. The two response traces share one gain within each frame; that gain may change between frames. Independent gains in the numerator and denominator do not cancel. Their ratio must be retained as another scale record. Thus "three readings" below means three *normalized invariant readings*, each constructed from two response traces and the known relative weight profile.

03 / IDENTIFIABILITY

What one comparison axis misses

The native matrices and transports are

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad U(q) = \mathrm{Cayley}(R, q), \quad V(p) = \mathrm{Cayley}(\iota K, p),$$

with $H = V^\dagger U^\dagger V U$. Here ι is the inherited native turn scalar. The matrix representation follows the native algebra; no physical Hilbert space or clock is added as a primitive.

C2. Separated inverse amplitude. Let $r = q/p$, $k = qp$, and define $Y(s,t) = 4/\sqrt{\chi(s,t)}$, using the positive root. Then

$$\boxed{Y(s,t) = rs^2 + \frac{1}{rs^2} + kt^2 + \frac{1}{kt^2}.}$$

Proof. For positive q', p', $4/\sqrt{\chi} = (1+q'^2)(1+p'^2)/(q'p')$. Substitute $q' = stq$, $p' = tp/s$ and expand. All denominators are positive. The denominator-cleared identity is checked by the native compiler. □

C3. Whole length-family blindness. At $t = 1$, the complete function $Y(s,1)$ depends on k only through $k + k^{-1}$. Hence every length-only scalar reading is unchanged under

$$(q,p) \longmapsto (1/p, 1/q), \qquad r \longmapsto r, \quad k \longmapsto k^{-1}.$$

For example, $(q,p) = (2,3)$ and $(1/3, 1/2)$ give the same invariant at every positive length ratio. The ordered loop matrices are generally different.

Even the unit-weight spectrum retains this ambiguity

For the four-link cost $L = (\mathbf{1} - S)^\dagger(\mathbf{1} - S)$, where S is the transported cycle shift, put

$$D(z) = z^4 + 8z^3 + 20z^2 + 16z + 4\chi.$$

Then, for $z > 0$,

$$\det(L + z\mathbf{1}_8) = D(z)^2, \qquad [(L + z\mathbf{1}_8)^{-1}]_{oo} = \frac{(z+2)(z^2+4z+2)}{D(z)}\mathbf{1}_2.$$

To see this, gauge three links to identity and set $c = z + 2$. The interior three-node chain has determinant $c(c^2-2)$ per local coordinate, end-diagonal inverse entry $(c^2-1)/[c(c^2-2)]$ and end-cross entry $1/[c(c^2-2)]$. Using $H + H^\dagger = 2(1-2\chi)\mathbf{1}_2$, its root Schur operator is $D(z)/[c(c^2-2)]\mathbf{1}_2$. The formulas follow; the determinant relation extends as a polynomial identity.

An alternative if orientation is observable

Keeping an oriented comparator permits the moment

$$-\frac{\mathrm{tr}[(R - \iota K)H]}{4\chi} = \frac{q+p}{2qp}(1 - qp).$$

Its sign distinguishes $k < 1$ from $k > 1$. It requires the comparator and loop to be retained in a common frame; under a joint unitary change of frame the trace is invariant. The protocol on the next page instead uses an independent source-energy comparison and scalar readings only.

04 / THE INVERSE PROTOCOL

Recover both transport ratios

C4. Three-reading reconstruction. Obtain the normalized invariants

$$\chi_0 = \chi(1,1), \qquad \chi_L = \chi(2,1), \qquad \chi_E = \chi(1,2),$$

and form $Y_0 = 4/\sqrt{\chi_0}$, $Y_L = 4/\sqrt{\chi_L}$, $Y_E = 4/\sqrt{\chi_E}$. Define

$$d_L = (Y_L - Y_0)/3, \qquad d_E = (Y_E - Y_0)/3.$$

The unique positive candidate is

$$\boxed{r = \frac{d_L + \sqrt{d_L^2 + 1}}{2}, \qquad k = \frac{d_E + \sqrt{d_E^2 + 1}}{2}, \qquad q = \sqrt{rk}, \quad p = \sqrt{k/r}.}$$

It is admitted only if the base-reading consistency condition holds:

$$\boxed{Y_0 = r + r^{-1} + k + k^{-1}.}$$

Proof. C2 gives $d_L = r - 1/(4r)$ and $d_E = k - 1/(4k)$. For positive x, y,

$$\left(x - \frac{1}{4x}\right) - \left(y - \frac{1}{4y}\right) = (x - y)\left(1 + \frac{1}{4xy}\right).$$

Thus the map is strictly increasing. Solving its quadratic gives exactly one positive root, as displayed. The consistency condition checks the remaining base equation. Conversely, that condition and the two difference equations reproduce all three input readings. The positive product and ratio uniquely determine q, p. □

This is a sufficient reconstruction protocol. No global minimality claim is made about every possible observation design.

Exact example: gains are unknown to the decoder

The synthetic cell uses $q = 2, p = 3$. Only the normalized outputs of the trace pairs are supplied to reconstruction; the hidden generating parameters are used afterwards for verification.

Setting (s,t)	$\operatorname{tr} K_{\text{bare}}$	$\operatorname{tr} K_{\text{eff}}$	χ	Y
$(1,1)$	12	864/625	144/625	25/3
$(2,1)$	20	92160/48841	9216/48841	221/24
$(1,2)$	14	64512/395641	9216/395641	629/24

All three frames use $\mathcal{N} = 2$; their hidden common gains are respectively $3, 5, 7/2$. The decoded differences and parameters are

$$d_L = \frac{7}{24}, \quad d_E = \frac{143}{24}, \quad r = \frac{2}{3}, \quad k = 6, \quad q = 2, \quad p = 3.$$

The native executable recovers these values exactly. Across the full test grid it reconstructs all 25 positive rational parameter pairs without numerical fitting.

Physical meaning of the second axis. A known source-energy ratio $t = 2$ must come from an independent source calibration. Merely doubling a native source-vector amplitude does not establish that physical energy doubled.

05 / A PREDICTION

Test the unused rectangle corner

C5. Rectangle identity. The separated response satisfies, for all positive s, t,

$$\boxed{Y(s,t) - Y(s,1) - Y(1,t) + Y(1,1) = 0.}$$

Each term of C2 depends on only one of the two axes, so the mixed difference cancels exactly. The native compiler also certifies its denominator-cleared polynomial form.

In particular, reconstruction from the three settings on page 4 predicts

$$Y(2,2) = Y_L + Y_E - Y_0, \qquad \boxed{\chi(2,2) = \frac{16}{(Y_L + Y_E - Y_0)^2}.}$$

For the exact example,

$$Y(2,2) = \frac{325}{12}, \qquad \chi(2,2) = \frac{2304}{105625}.$$

A direct native construction of the fourth loop returns this same rational number. The fourth value is not used by the decoder.

What a mismatch would tell us

A nonzero rectangle residual rejects the stated combination of assumptions: fixed calibration, known independent scale ratios, the two monomial transport laws, the cycle model and the response normalization. By itself it does not identify which assumption failed.

For exact records, define the diagnostic

$$\mathcal{R}(s,t) = Y(s,t) - Y(s,1) - Y(1,t) + Y(1,1).$$

An experimental use would also need uncertainty propagation from the trace measurements and the scale references. This packet supplies an exact model test; it does not choose a statistical tolerance or claim noise robustness.

Negative controls that prevent overinterpretation

Deliberate change	Observed consequence in exact replay
Inconsistent three-reading input	Rejected by the base-reading consistency equation.
Change q by 11/10 only at the fourth setting	The held-out prediction fails.
Double only the effective-response detector gain	Normalized χ is wrong; an independent scale record is needed.
Refine the cycle but retain $\mathcal{N} = 2$	The invariant is misnormalized. The correct refined value is $\mathcal{N} = 4$.

Four successful settings do not establish a global law

Consider the calibration correction

$$\alpha(s,t) = 1 + \frac{(s-1)(s-2)(t-1)(t-2)}{100}$$

applied multiplicatively to $q(s,t)$. It equals one at all four tested corners, yet changes the response at $(3,3)$ for the example. This is an explicit counterexample to inferring the whole comparison law from one successful rectangle.

The practical next test is therefore a collection of independently chosen rectangles with independently calibrated source and length ratios. Their purpose is to challenge the adapter, not to refit every corner separately.

06 / EXACT GEOMETRY

The comparison circle appears

C6. Unique relative balance point. Since $x + x^{-1} \geq 2$ for positive x, C2 gives $Y \geq 4$ and $\chi \leq 1$. Equality holds precisely when $rs^2 = 1$ and $kt^2 = 1$. Therefore

$$\boxed{s_*^2 = \frac{1}{r}, \qquad t_*^2 = \frac{1}{k}, \qquad q(s_*, t_*) = p(s_*, t_*) = 1.}$$

This is the previously selected nontrivial compatible cell: $H = -\mathbf{1}_2$. In the exact example, $s_*^2 = 3/2$ and $t_*^2 = 1/6$.

Center the comparison ratios by $\sigma = s/s_*$, $\tau = t/t_*$. The two transports become

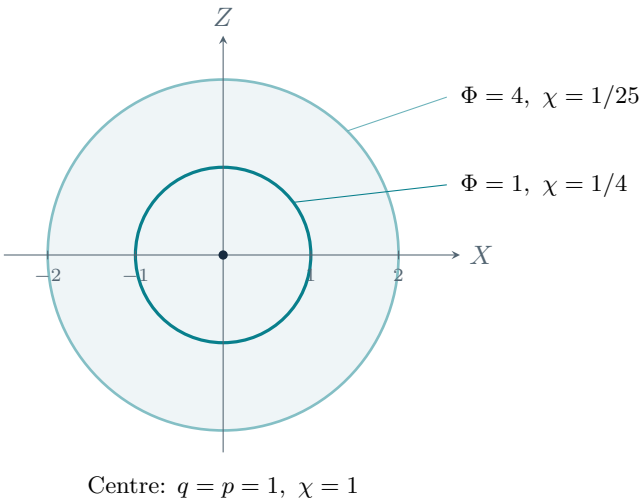
$$q = \sigma\tau, \qquad p = \tau/\sigma.$$

Introduce the signed, dimensionless coordinates

$$X = \frac{\sigma - \sigma^{-1}}{2}, \qquad Z = \frac{\tau - \tau^{-1}}{2}.$$

Both maps are bijective from positive ratios to real coordinates: for instance $\sigma = X + \sqrt{1 + X^2}$. Using $(a - a^{-1})^2 = a^2 + a^{-2} - 2$ in C2 proves

$$\boxed{\Phi := \frac{1}{\sqrt{\chi}} - 1 = X^2 + Z^2, \qquad \chi = (1 + X^2 + Z^2)^{-2}.}$$



Exact level sets in the centered length/source-energy comparison plane.

The maximum of χ is unique under this declared comparison family. No physical attraction toward the centre, universal minimum length, spacetime circle or time evolution follows from that extremum alone. All of the geometry above comes from retained scale comparisons.

07 / LOCAL RESOLUTION

The balanced cut can still be probed

At the selected point $q = p = 1$, the central invariant has vanishing first-order change in either parameter. Reading only this central maximum is therefore locally insensitive at first order. The off-centre comparisons retain two independent directions.

C7. Full-rank two-probe sensitivity. Consider the pair

$$F(q,p) = \begin{pmatrix} \chi(2q, p/2) \\ \chi(2q, 2p) \end{pmatrix}.$$

Its first-order coefficient matrix at $q = p = 1$ is

$$J = \begin{pmatrix} -d & d \\ -d & -d \end{pmatrix}, \qquad d = \frac{1536}{3125}, \qquad \det J = \frac{4718592}{9765625} > 0.$$

The rows are the length and source-energy probes; the columns are the initial q, p parameters. Their perturbations can therefore be separated to first order.

A native first-order proof

Declare an auxiliary self-adjoint jet token j with $j^2 = 0$. This is a quotient for recording first-order variation, not a physical time parameter. For a perturbed argument, form the rational numerator and denominator of χ and verify the identity

$$\text{Numerator}(j) = \left(\frac{256}{625} \pm \frac{1536}{3125} j\right) \text{Denominator}(j) \pmod{j^2}.$$

The appropriate signs are exactly the four entries of J. The native compiler proves and replays these four denominator-cleared equalities. The denominator has a nonzero constant coefficient, so the rational expression is well-defined in the jet quotient.

Probe	q coefficient	p coefficient	Direction resolved
Length: $(s,t) = (2,1)$	$-d$	d	Opposing variation
Source energy: $(s,t) = (1,2)$	$-d$	$-d$	Common variation

The observer must allow the comparison

The inherited compatibility theorem for two fixed, independent observer cuts selects $q = p = 1$ in the positive nontrivial sector. The present probes deliberately visit other loop values. Their observer defects must remain in the record.

If one instead demanded exact two-cut compatibility at every setting, every admitted probe would stay at $q = p = 1$ and this inverse protocol would lose its comparison freedom. Thus the development extends the observation protocol; it does not replace the earlier selection theorem.

What rank two establishes

It establishes local first-order identifiability in the declared model. It does not establish an optimal experimental design, a bound on uncertainty, a stable inverse over arbitrary ranges, or a physical coupling law. Those questions require an actual source and readout adapter.

The global positive-parameter reconstruction on page 4 is established by its explicit equations and consistency condition, separately from this local sensitivity calculation.

08 / PHYSICAL CALIBRATION

What becomes measurable now

Retain the physical dictionary from the previous packet:

$$A = \hbar c, \qquad B = G/c^4, \qquad q = \alpha \frac{E\ell}{A}, \qquad p = \beta \frac{BE}{\ell}.$$

Here $\ell > 0$ is an operational length, $E > 0$ is the source-energy scale, and $\alpha, \beta > 0$ are dimensionless calibration records. A has units energy times length; B has units length divided by energy. The symbol α is not the fine-structure constant.

Holding A, B, α, β fixed while taking $\ell \mapsto s\ell$, $E \mapsto tE$ gives exactly the comparison law used in this note. This is a declared physical adapter, not a law forced by the operator algebra.

C8. Calibration recovery with references. If the reference readings ℓ, E, A, B are independently supplied, the recovered q, p determine

$$\boxed{\alpha = \frac{qA}{E\ell}, \qquad \beta = \frac{p\ell}{BE}.}$$

For the illustrative exact references $(\ell, E, A, B) = (2, 3, 5, 7)$ and decoded $(q, p) = (2, 3)$, this gives $\alpha = 5/3$ and $\beta = 2/7$. These are synthetic reference values in a consistent unit chart, not empirical measurements of physical constants.

The relative balance acquires a physical address

The reconstructed r, k locate

$$\ell_*^2 = \frac{\ell^2}{r} = \frac{\beta}{\alpha} AB, \qquad E_*^2 = \frac{E^2}{k} = \frac{A}{\alpha\beta B}.$$

With the stated conventional dictionary this reads

$$\ell_*^2 = \frac{\beta}{\alpha} \frac{\hbar G}{c^3}, \qquad E_*^2 = \frac{1}{\alpha\beta} \frac{\hbar c^5}{G}.$$

The earlier conditional Planck correspondence now has an inverse protocol for its dimensionless factors when physical references exist. Nothing here forces $\alpha = \beta = 1$.

The remaining absolute ambiguity is explicit

At fixed A, B, for arbitrary positive u, v, the transformation

$$\ell \mapsto u\ell, \quad E \mapsto vE, \quad \alpha \mapsto \frac{\alpha}{uv}, \quad \beta \mapsto \frac{u\beta}{v}$$

preserves q, p and every relative probe $\chi(s, t)$. Therefore relative invariant readings alone cannot determine all four absolute records. Independent physical references remove this ambiguity; more copies of the same relative invariant do not.

Scope. No time variable enters this protocol. The source energy E is not identified with an eigenvalue of the graph cost. The model does not derive an Einstein field equation, an empirical value of G, or a complete quantum theory of gravity. It supplies a concrete calibration and falsification target for the proposed clock-free construction.

The conventional physical definitions are inherited from the previous packet's NIST CODATA source, P. J. Mohr et al., *Rev. Mod. Phys.* **97**, 025002 (2025), Table XXXIII. They are a dictionary, not evidence for the new comparison law.

09 / REPRODUCIBLE EVIDENCE

Proofs, checks and boundaries

Evidence	What is actually checked
6 symbolic identities	Separated inverse amplitude, rectangle cancellation, the two axis differences, whole length-family duality, and the centered circle polynomial.
4 symbolic jet identities	The four first-order sensitivity coefficients at the selected cut.
10 serialized replays	Each compiler certificate is serialized and replayed in its declared quotient.
610 exact checks	Includes presentation admission, certificate/replay checks, native rational and matrix assertions. This is not a count of independent experiments.
25 parameter pairs	$q, p \in \{1/3, 1/2, 1, 2, 3\}$; trace normalization, data-only reconstruction, held-out predictions, duality and spectral response.
16 centered cases	$\sigma, \tau \in \{1/2, 1, 2, 3\}$; circle identity and the unique centre on this grid.
10 negative controls	Ambiguities, inconsistent inputs, forbidden float fallback, coupling changes, gain mismatch, wrong refinement normalization and remaining absolute calibration freedom.

The general results are supported by the proofs in this note and the specified symbolic identities. Finite examples check the implementation; they do not replace the general proofs. In particular, the graph-normalization and full resolvent statements have written algebraic proofs and exact matrix spot checks; they are not among the ten polynomial compiler certificates.

Arithmetic and source continuity

All mathematical arithmetic is executed by the unchanged native engine and compiler included in the parent packet. The rational-root adapter uses native exact arithmetic and accepts only perfect rational squares. It rejects an irrational root instead of silently approximating it. General positive-root statements in the proofs use the inherited real completion; an executable decoder for arbitrary noisy or irrational observations is not supplied here.

Python 3.12 performs file-integrity checks and launches Node. No second scalar engine, floating-point fit, SymPy calculation or numerical eigensolver is used. The auxiliary first-order jet is an explicitly declared native quotient.

Canonical source continuity is pinned to Recognition-Kernel-Framework commit

`e8e3089745bf7dc0532d01c62a5f660f5658c6f0`.

The included engine files, inherited manifests and upstream source blobs are checked before the new replay. The parent directory contains the complete preceding physics-cubit packet, including its earlier quantisation and lattice packets. Their files are unchanged.

Reproduce from the source archive

Extract the archive and run, from its top-level folder:

```
python3.12 verify.py
```

Expected result: `PASS_NATIVE_SCALE_COMPASS`, with the counts above. The verifier also compares the generated result byte-for-byte with `SCALE_COMPASS_RESULT.json`. To rebuild the note, run `pdflatex` twice on `Clock_Free_Scale_Compass.tex`.

Next research target. Supply an independently calibrated source/readout adapter and test the rectangle relation across several unused settings. This is the concrete step that can distinguish a successful physical mapping from an internally consistent operator model.